

# Lecture 1:

## Superposition and quantum interference

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# Quantum computation: The *motto*

Information is a physical entity. Any computation is a physical process

Thus, any abstract computational model has always to reflect **what we know about reality**, and **quantum theory** is our **current best tool** in this road.

## Recall: The **physical** Church-Turing thesis

The class of functions computable in finite time by a physical device can be computed by a Turing machine.

which a statement about the **physical universe**, while the Church-Turing thesis — *The class of functions computable by a Turing machine corresponds exactly to the class of functions which can be described by an algorithm* — is a claim about what counts as an algorithm

# Information is physical

How to represent information in a computational device?

Three basic ways ...

- **Binary** ... in a **two-state** device
- **Probabilistic** ... in a similar way but with **uncertainty**
- **Quantum** ... in a **quantum** device

## Binary information

Computational / information states based on Boolean values 0 and 1 which can be represented by **vectors**:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

If rows are labelled from 0 onwards, the presence of 1 in a cell identifies the number represented by the vector.

Larger state spaces are built with the (Kronecker) **tensor** product:

$$\begin{bmatrix} p_0 \\ p_1 \end{bmatrix} \otimes \begin{bmatrix} q_0 \\ q_1 \end{bmatrix} = \begin{bmatrix} p_0 \begin{bmatrix} q_0 \\ q_1 \end{bmatrix} \\ p_1 \begin{bmatrix} q_0 \\ q_1 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} p_0 q_0 \\ p_0 q_1 \\ p_1 q_0 \\ p_1 q_1 \end{bmatrix}$$

# Binary information

## Examples

$$|00\rangle = |0\rangle \otimes |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$|4\rangle = |100\rangle = |1\rangle \otimes |0\rangle \otimes |0\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

# Binary information

## Operations as matrices

$$I(x) = x$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$X(x) = \neg x$$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\underline{1}(x) = 1$$

$$\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\underline{0}(x) = 0$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$I|0\rangle = |0\rangle \quad I|1\rangle = |1\rangle$$

$$X|0\rangle = |1\rangle \quad X|1\rangle = |0\rangle$$

$$\underline{1}|0\rangle = |1\rangle \quad \underline{1}|1\rangle = |1\rangle$$

$$\underline{0}|0\rangle = |0\rangle \quad \underline{0}|1\rangle = |0\rangle$$

# Binary information

## Composition

Sequential composition: **matrix multiplication**

Parallel composition: **Kronecker product**  $\otimes$

$$M \otimes N = \begin{bmatrix} M_{1,1}N & \cdots & M_{1,n}N \\ \vdots & & \vdots \\ M_{m,1}N & \cdots & M_{m,n}N \end{bmatrix}$$

for example

$$X \otimes \underline{1} \otimes I |101\rangle = X \otimes \underline{1} \otimes I (|1\rangle \otimes |0\rangle \otimes |1\rangle) = X|1\rangle \otimes \underline{1}|0\rangle \otimes I|1\rangle = |011\rangle$$

## Probabilistic information

... the system is **always in some well defined state**, even if we do not know which:

**State**: is a **vector of probabilities** in  $\mathcal{R}^n$

$$[p_0 \cdots p_n]^T \text{ such that } \sum_i p_i = 1$$

which expresses **indeterminacy** about the system's exact physical configuration

**Operator**: is a **double stochastic** matrix where  $M_{i,j}$  specifies the probability of evolution from state  $j$  to  $i$



# Quantum information is a different story

A quantum state holds the information of **both** possible classical states:



A unit of information lives in a 2-dimensional **complex** vector space:

$$|\nu\rangle = \alpha|0\rangle + \beta|1\rangle$$

and thus possesses a **continuum of possible values**, so potentially, can store lots of classical data.

# Quantum information is a different story

However, all this potential is **hidden**:

when **observed**  $|v\rangle$  **collapses into a classic state**:  $|0\rangle$ , with probability  $\|\alpha\|^2$ , or  $|1\rangle$ , with probability  $\|\beta\|^2$ .

(Recall:  $\|\alpha\| = \sqrt{\alpha\bar{\alpha}}$  for a complex  $\alpha$ )

The outcome of an observation is **probabilistic**, which calls for a restriction to **unit** vectors, i.e. st

$$\|\alpha\|^2 + \|\beta\|^2 = 1$$

to represent quantum states.

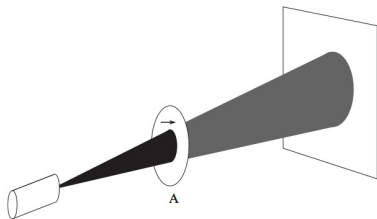
## Quantum information is a different story

This quantum state is **not** a probabilistic mixture: it is **not** true that the state is really either  $|0\rangle$  or  $|1\rangle$  and we just do not happen to know which.

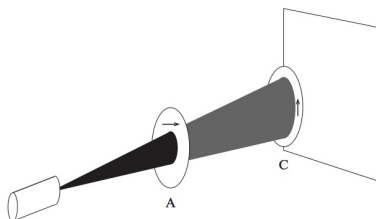
**Amplitudes are not real numbers** (e.g. probabilities) that can only increase when added, but **complex** so that they can **cancel each other or lower their probability**, thus capturing a fundamental **quantum resource**:

**interference**

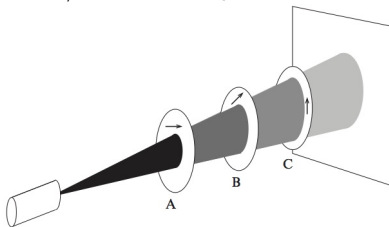
## Quantum information: An experiment with a photon



$|0\rangle$  - horizontal polarization



$|1\rangle$  - vertical polarization



(from [Reifell & Polak, 2011])

# Quantum information: An experiment with a photon

## An explanation for a *single* photon experiment

- The photon's polarization state is modelled by a unit vector, for example the following **linear combination of  $|0\rangle$  and  $|1\rangle$** :

$$\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

which corresponds to a polarization of 45 degrees.

- On passing a polaroid the photon will be absorbed or leave the polaroid with its polarization aligned with the polaroid's axis.
- The probability to go through the polaroid is the **square of the magnitude** of the amplitude of its polarization in the direction of the polaroid's axis.

## Quantum information: An experiment with a photon

- Polarization of polaroid B is  $|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$
- This vector, together with  $|-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$  forms a **basis** for the 2-dimensional vector space
- of which  $\{|0\rangle, |1\rangle\}$  is another one (the so-called **computational basis**).
- Expressing  $|0\rangle$  in terms of  $|+\rangle$  and  $|-\rangle$  yields

$$|0\rangle = \frac{1}{\sqrt{2}}|-\rangle + \frac{1}{\sqrt{2}}|+\rangle$$

which explains why a visible effect appears in the wall:

the photon goes through C with 50% of probability  
(i.e.  $\|\frac{1}{\sqrt{2}}\|^2 = \frac{1}{2}$ ).

## Why aren't probabilities enough?

Computation is always a physical process

That's our *motto*!

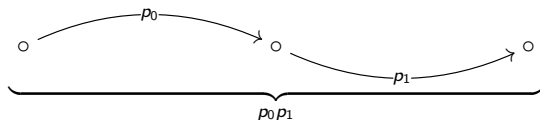
In several cases, the language of **probability theory** can describe the actual **physical evolution of a system**, i.e.

- **Physics** identify the system's structure and assigns numerical probabilities to elementary transition steps.
- **Probability theory**, i.e. the Kolmogorov axioms, ensure mathematical consistency and helps in calculating probabilities along paths of evolution.

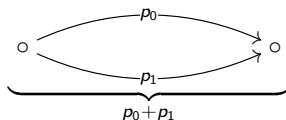
but **not always**!

# Probabilistic systems, probabilistic computation

## Sequential paths



## Alternative, mutually exclusive paths



(Kolmogorov (1903-1987) [additivity axiom](#))



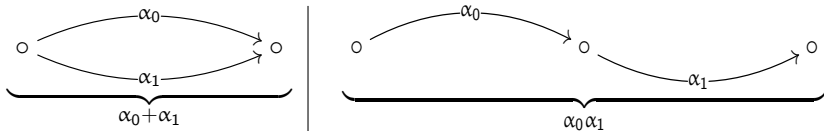
# Quantum systems, quantum computation

Many common quantum phenomena, however, cannot be described this way, but are accommodated by a **modified 'probability theory'**:

Transitions are labelled by **complex** numbers, called their **amplitudes**, whose **norms squared** are interpreted as **transition probabilities** through

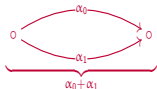
$$\text{Born's rule } p = \|\alpha\|^2$$

(for Max Born, 1882-1970)



# Quantum systems, quantum computation

Let's compute the total probability in



$$\begin{aligned} p &= \|\alpha_0 + \alpha_1\|^2 \\ &= \overline{(\alpha_0 + \alpha_1)}(\alpha_0 + \alpha_1) \\ &= (\overline{\alpha_0} + \overline{\alpha_1})(\alpha_0 + \alpha_1) \\ &= \|\alpha_0\|^2 + \|\alpha_1\|^2 + \overline{\alpha_0}\alpha_1 + \alpha_0\overline{\alpha_1} \\ &= p_0 + p_1 + \|\alpha_0\| \|\alpha_1\| \left( e^{i(\varphi_1 - \varphi_0)} + e^{-i(\varphi_1 - \varphi_0)} \right) \\ &= p_0 + p_1 + \underbrace{2\sqrt{p_0 p_1} \cos(\varphi_1 - \varphi_0)}_{\text{interference}} \end{aligned}$$

(expressing  $\alpha_j$  in polar form  $\|\alpha_j\| e^{i\theta_j}$   
and resorting to  $e^{i\theta} + e^{-i\theta} = 2 \cos \theta$ )

## Quantum systems, quantum computation

$$p = p_0 + p_1 + \underbrace{2\sqrt{p_0 p_1} \cos(\varphi_1 - \varphi_0)}_{\text{interference}}$$

- The total probability is the sum of the probabilities of the individual transitions **modified by the interference term**.
- Depending on term  $\varphi_1 - \varphi_0$  the interference can be either **negative** or **positive**.
- The important quantity is the **relative phase  $\varphi_1 - \varphi_0$**  rather than individual  $\varphi_0, \varphi_1$ .
- If the system's evolution depends only on that difference then the system **must have, somehow, experienced both paths**.

# Quantum systems, quantum computation

Indeed, the **probabilistic** assumption that one or the other transition occurs, or that the system presents one or the other configuration, but **we just do not know which one**, is inconsistent with many experiments:

Typically, a quantum state is a **superposition** of basic states, i.e. states that, mathematically, form a **basis** of the vector space in which such states live — e.g.  $\{|e_i\rangle \mid i = 1 \cdots n\}$

$$|\varphi\rangle = \sum_i \alpha_i |e_i\rangle$$

Recall



# Superposition in action: A random number generator

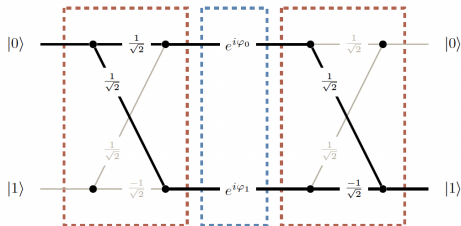
- Prepare quantum state

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

- Measure  $|+\rangle$  in the computational basis to obtain either 0 or 1 with equal probability ( $\|\frac{1}{\sqrt{2}}\|^2 = 0.5$ )

This algorithm produces a perfect random number even though no randomness has been used inside it.

## Ramsey interference experiment



(from [Ekert *et al*, 2024])

Atoms are sent through two separate resonant interaction zones (which attempt to commute the atom between ground and excited states), separated by an intermediate dispersive interaction zone (which performs a phase shift).

The atoms are subsequently measured and found to be in one of the two basic energy states labeled as  $|0\rangle$  and  $|1\rangle$ .

# What is the probability of an atom going from $|0\rangle$ to $|1\rangle$ ?

First compute the **amplitudes**:

$$\begin{aligned}A_{10} &= \frac{1}{\sqrt{2}} e^{i\varphi_0} \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} e^{i\varphi_1} \frac{-1}{\sqrt{2}} \\&= \frac{1}{2} (e^{i\varphi_0} - e^{i\varphi_1}) \\&= \frac{1}{2} \left( e^{i\frac{\varphi_0+\varphi_1}{2}} e^{i\frac{\varphi_0-\varphi_1}{2}} - e^{i\frac{\varphi_0+\varphi_1}{2}} e^{-i\frac{\varphi_0-\varphi_1}{2}} \right) \\&= \frac{1}{2} e^{i\frac{\varphi_0+\varphi_1}{2}} \left( e^{i\frac{\varphi_0-\varphi_1}{2}} - e^{-i\frac{\varphi_0-\varphi_1}{2}} \right) \\&= \frac{1}{2} e^{i\frac{\varphi_0+\varphi_1}{2}} \left( 2i \sin \frac{\varphi_0 - \varphi_1}{2} \right) \\&= -ie^{i\frac{\varphi_0+\varphi_1}{2}} \sin \frac{\varphi_1 - \varphi_0}{2}\end{aligned}$$

resorting to  $x = \frac{x+y}{2} + \frac{x-y}{2}$   
and Euler formula  $e^{i\varphi} = \cos \varphi + i \sin \varphi$

# What is the probability of an atom going from $|0\rangle$ to $|1\rangle$ ?

Then the probability:

$$\begin{aligned}
 P_{10} &= \|A_{10}\|^2 \\
 &= \left\| -ie^{i\frac{\varphi_0 + \varphi_1}{2}} \sin \frac{\varphi_1 - \varphi_0}{2} \right\|^2 \\
 &= \left\| \sin \frac{\varphi_1 - \varphi_0}{2} \right\|^2 \\
 &= \frac{1}{2} - \underbrace{\frac{1}{2} \cos(\varphi_1 - \varphi_0)}_{\text{interference}}
 \end{aligned}$$

resorting to  $\cos 2\theta = 1 - 2 \sin^2 \theta$



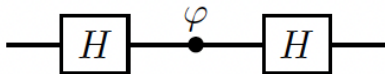
## A bulk calculation ...

The results of these calculations for all pairs begin-end states can be expressed as follows:

- the effect of each interaction is described by a **matrix of transition amplitudes**, and
- one resorts to **matrix multiplication** to compose the sequence of independent interactions.

$$\begin{aligned}
 A &= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} e^{i\varphi_0} & 0 \\ 0 & e^{i\varphi_1} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix} \\
 &= e^{i\frac{\varphi_0 + \varphi_1}{2}} \begin{bmatrix} \cos \frac{\varphi}{2} & -i \sin \frac{\varphi}{2} \\ -i \sin \frac{\varphi}{2} & \cos \frac{\varphi}{2} \end{bmatrix} \quad \text{making } \varphi = \varphi_1 - \varphi_0 \\
 &\equiv \begin{bmatrix} \cos \frac{\varphi}{2} & -i \sin \frac{\varphi}{2} \\ -i \sin \frac{\varphi}{2} & \cos \frac{\varphi}{2} \end{bmatrix} \quad \text{ignoring global phase, to discuss later ...}
 \end{aligned}$$

## My first quantum circuit



- A **wire** represents a two-dimensional **state** (a **qubit**)
- Three **gates** describing quantum **operations**:

$$\underbrace{H = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix}}_{\text{Hadamard gate}} \quad \text{and} \quad \underbrace{P_\varphi = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\varphi} \end{bmatrix}}_{\text{Phase shift gate}}$$

A simple matrix multiplication yields

$$A = HP_\varphi H = \begin{bmatrix} \cos \frac{\varphi}{2} & -i \sin \frac{\varphi}{2} \\ -i \sin \frac{\varphi}{2} & \cos \frac{\varphi}{2} \end{bmatrix} = \begin{bmatrix} A_{00} & A_{01} \\ A_{10} & A_{11} \end{bmatrix}$$

## My first quantum circuit

which, expressed in a functional way, gives

$$A|0\rangle = \cos \frac{\varphi}{2} |0\rangle + -i \sin \frac{\varphi}{2} |1\rangle$$

$$A|1\rangle = -i \sin \frac{\varphi}{2} |0\rangle + \cos \frac{\varphi}{2} |1\rangle$$

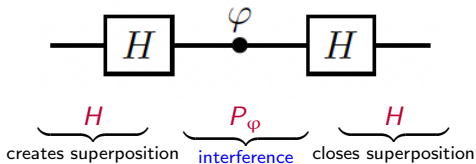
as read from

$$A = HP_{\varphi}H = \begin{bmatrix} \cos \frac{\varphi}{2} & -i \sin \frac{\varphi}{2} \\ -i \sin \frac{\varphi}{2} & \cos \frac{\varphi}{2} \end{bmatrix}$$

Clearly, for  $\varphi = 0$ , i.e. in the absence of any phase shift,  $A|0\rangle = |0\rangle$  and  $A|1\rangle = |1\rangle$ , leading to conclude that

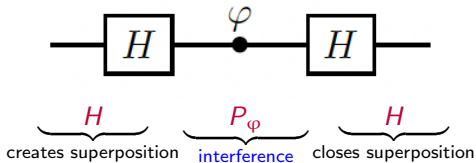
$$HH = Id$$

## The golden pattern



- $H$  creates/closes a **uniform superposition**: it is the source of a natural parallelism,
- but the **crucial** role in controlling interference is located in  $P_\varphi$ .

## Lessons learnt for what follows



generalizes to a basic **recipe** in quantum algorithms (to be discussed).

### What is quantum computation?

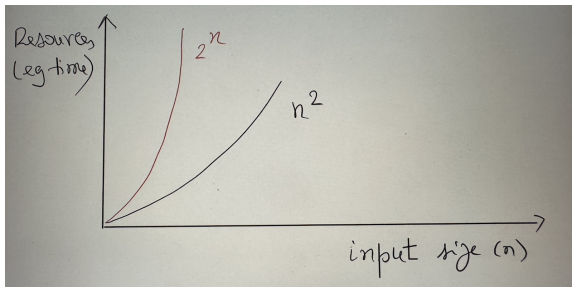
A complex multi-particle quantum **interference involving many computational paths** through a computing device.

Its challenge is **to shape quantum interference** through a sequence of computational steps, enhancing probabilities of the **correct** outputs and suppressing probabilities of the **wrong** ones.

# Superposition and interference what for?

Quantum computing has nothing to say on the boundaries of **computability**, but reduces the **complexity** of algorithms to address several problems.

## The limits of practicality



## Paying a visit to old acquaintances

**EXP** Problems requiring exponential time to be solved

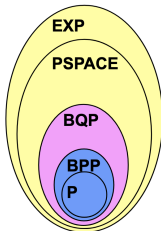
**PSPACE** Problems requiring polynomial space to be solved

**BQP** Problems solved in quantum machines in bounded error polynomial time

**BPP** Problems solved in probabilistic machines in bounded error polynomial time

**P** Problems solved in deterministic classical machines in polynomial time

$$P \subseteq BPP \subseteq BQP \subseteq PSPACE \subseteq EXP$$

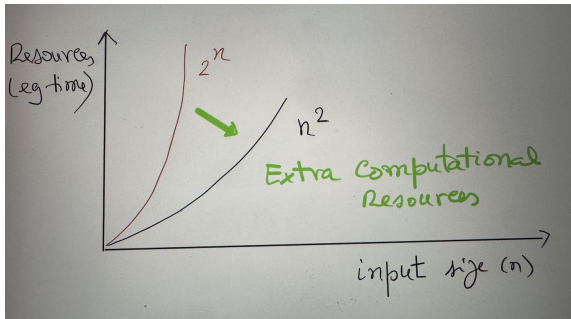


# Superposition and interference as computational resources

A problem does not become **tractable** by increasing computational power:  
more of the same keeps similar circumstances.

New ideas & new resources make a difference

The key: Quantum effects as computational resources





# Decoherence

The laws of quantum computation just described assume a **perfectly isolated system**.

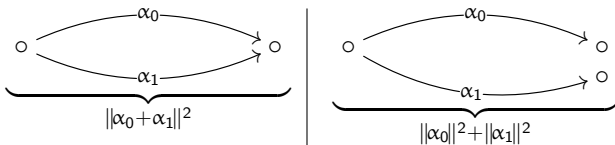
- In practice, any complex quantum system cannot avoid some spurious **interactions with the environment**.
- Thus, when computing the system's evolution, one must take into account not only its internal configurations but also those of its environment.

This is known as **decoherence** and explains why

- we hardly see quantum interference on a daily basis
- and isolation is a main technological challenge in the design/construction of quantum computing devices.

## Two scenarios

- **Perfect isolation:** the environment does not hold any physical record of the path taken to produce an output.
- **Total decoherence:** the system produces an output but the environment has a record of the path taken: there are no alternative ways to reach that output, but a single path to each of the following situations (joint states):
  1. The output was produced, but the environment knows the path labelled by  $\alpha_0$  was taken,
  2. or similarly for  $\alpha_1$ .



## Two scenarios

- **Perfect isolation**: amplitudes are added and generate a probability for the output to be produced, taking both paths in superposition
- **Total decoherence**: probabilities are computed for each path and summed as such — **interference** provided by superposition is **lost**.

### The general scenario

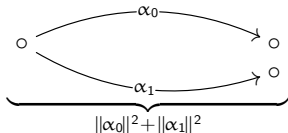
Interference is limited by a **visibility** parameter —  $v$  — ranging from

- **0**: total decoherence; no interference
- **1**: no decoherence; full interference

## The effect of decoherence

$$p_0 + p_1 + 2\mathfrak{v}\sqrt{p_0 p_1} \cos(\phi_1 - \phi_0)$$

Parameter  $\mathfrak{v}$  quantifies the **degree of distinguishability between the disrupted outputs**: *the more the environment knows the less interference is observed more classic becomes the overall computation.*



### Fact

Decoherence destroys (quantum) interference.